

SEDAR 40 Atlantic Menhaden

Data/Assessment Workshops

Research recommendations are broken down into two categories: data and modeling. While all recommendations are high priority, the first recommendation is the highest priority. Each category is further broken down into recommendations that can be completed in the short term and recommendations that will require long term commitment.

Annual Data Collection

Short term (next 3-6 years):

1. Continue current level of sampling from bait fisheries, particularly in the Mid-Atlantic and New England. Analyze sampling adequacy of the reduction fishery and effectively sample areas outside of that fishery (e.g., work with industry and states to collect age structure data and biological data outside the range of the fishery).
2. Ageing:
 - a. Conduct ageing validation study (e.g., scale : otolith comparison), making sure to sample older age classes. Use archived scales to do radio isotope analysis.
 - b. Ageing precision: conduct an ageing workshop to assess precision and error among readers (currently planned for January 2015).
3. Conduct a comprehensive fecundity study.
4. Place observers on boats to collect at-sea samples from purse-seine sets, or collect samples at dockside during vessel pump-out operations (as opposed to current top of hold sampling) to address sampling adequacy.
5. Investigate relationship between fish size and school size in order to address selectivity (specifically addressing fisher behavior related to harvest of specific school sizes).
6. Investigate relationship between fish size and distance from shore (addressing selectivity).
7. Evaluate alternative fleet configurations for the removal and catch-at-age data.

Long term (6+ years):

1. Develop a menhaden specific coastwide fishery-independent index of adult abundance at age. One possible methodology is an air spotter survey complemented with ground truthing for biological information (e.g., size and age composition). In all cases, a sound statistical design is essential (involving statisticians in the development and review of the design; some trial surveys may be necessary). **[Highest Priority]**
2. Conduct studies on spatial and temporal dynamics of spawning (how often, how much of the year, batch spawning, etc.)
3. Conduct studies on productivity of estuarine environments related to recruitment.
4. Investigation of environmental covariates related to recruitment.

Assessment Methodology

Short term (3-6 year):

1. Conduct management strategy evaluation (MSE). **[Highest Priority]**
2. Conduct multi-objective decision analysis (MODA). **[Highest Priority]**

3. Continue to develop an integrated length and age based model (e.g., SS3).
4. Continue to improve methods for incorporation of natural mortality (e.g., multi-species statistical catch-at-age model). See Appendix E.

Long term (6+ years):

1. Develop a seasonal spatially-explicit model, once sufficient age-specific data on movement rates of menhaden are available.

Review Workshop

The Technical Committee developed a set of “Research and Modeling Recommendations” that were categorized by time frame (short versus long term) and research type (data collection versus assessment methodology). The panel generally agreed with the TC’s recommendations. There was strong agreement that developing a coast-wide fishery-independent index of abundance-at-age is the top priority for data collection. Related to this was a suggestion that collection of age composition data for the existing fishery independent surveys should also be a high priority. The AT noted that this was reflected in the existing recommendations under item 1 in the short-term data priorities: “work with industry and states to collect age structure data and biological data outside the range of the fishery”. Given the challenges, discussed elsewhere in this report, of using index length-frequency data to inform index selectivity-at-age in the model, the panel concluded that having direct estimates of survey age composition would be a very valuable addition to the assessment data.

The panel also agreed that conducting a Management Strategy Evaluation to evaluate the performance of alternative harvest strategies and possibilities for reference points should be a high priority for the immediate future. Ideally the MSE should be informed by a structured Decision Analysis process (also listed as a research recommendation) that would both inform the MSE with respect to management objectives and options, and provide an opportunity for the MSE to be transparent for both stakeholders and decision makers.

The panel expressed some reservations about the recommendation to “develop an integrated length and age based model” and greater reservations about the recommendation to “develop a seasonal spatially-explicit model, once sufficient age-specific data on movement rates of menhaden are available”. If the AT pursues the former, it may be more fruitful to adapt the BAM to integrate length and age than to use an alternative modeling platform, given the obvious expertise the AT has with the BAM. Regarding the latter, the panel cited previous experience with numerous challenges associated with developing spatial assessment models that explicitly incorporate movement, implying that the benefits (in terms of informing menhaden management) of pursuing this modeling strategy might not outweigh the costs (in terms of scientific effort).

The panel noted that two aspects of modeling are currently conducted in advance of the main BAM model:

- 1) Growth modeling.
- 2) Scaling of natural mortality based on tag data.

The panel suggests investigating the potential for including these aspects of the analysis as part of the assessment model. The model currently uses growth (length-at-age) as a basis for several aspects of the model. It may be possible to estimate selectivity-at-age using age data for the NAD and SAD survey indices, but if that is not the case and length composition data continue to be used in the assessment model, estimation of growth could be integrated in the assessment model.

If time-invariant mortalities are to be considered as part of future modeling, consideration should be given to estimating natural mortality in the assessment model, informed by the tagging data that are currently used externally. This would help to integrate the estimation process.